

SUNDANCE 8 NNW, WY, USA

WMO: 755750

Lat: 44.5169N

Lon: 104.4363W

Elev: 5792

StdP: 11.87

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 94088

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.4	-6.3	-17.0	2.7	-5.6	-12.4	3.5	-3.1	18.5	29.9	16.9	29.4	4.5	N/A	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	82.5	58.6	79.6	58.3	76.9	58.0	63.9	74.1	62.1	73.1	60.6	72.4	5.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	96.8	69.5	58.2	90.0	68.0	56.1	83.4	66.1	32.4	74.2	30.9	73.4	29.7	72.5	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.4	12.7	DB	-18.2	86.8	5.9	2.6	-22.5	88.7	-25.9	90.2	-29.3	91.7	-33.6	93.6	(4)
(5)			WB	-18.1	67.3	6.0	2.3	-22.4	68.9	-26.0	70.2	-29.3	71.5	-33.7	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	23.0	20.7	31.6	37.0	46.1	56.8	64.4	63.0	55.6	41.4	31.6	21.6	(6)
(7)		DBStd	18.55	12.10	12.43	11.75	10.02	9.67	7.84	6.24	6.92	9.79	11.30	12.10	12.82	(7)
(8)		HDD50	4662	838	820	576	406	193	24	1	3	57	308	557	879	(8)
(9)		HDD65	8868	1303	1240	1036	841	590	265	86	122	306	734	1001	1344	(9)
(10)		CDD50	1445	0	0	5	14	71	229	449	405	224	41	7	0	(10)
(11)		CDD65	175	0	0	0	0	3	19	69	59	23	2	0	0	(11)
(12)		CDH74	1280	0	0	0	0	14	153	523	443	141	6	0	0	(12)
(13)		CDH80	217	0	0	0	0	1	28	101	76	11	0	0	0	(13)
(14)	Wind	WSAvg	6.0	7.4	6.7	6.5	6.2	5.8	5.3	4.6	4.7	5.3	6.1	6.7	7.0	(14)
(15)	Precipitation	PrecAvg	18.5	0.6	0.7	1.1	2.2	3.1	3.0	2.1	1.5	1.2	1.6	0.8	0.6	(15)
(16)		PrecMax	24.1	1.5	1.4	2.2	3.7	6.5	6.7	3.7	3.0	3.1	5.3	3.0	1.7	(16)
(17)		PrecMin	13.5	0.2	0.2	0.5	0.9	1.2	0.6	0.6	0.5	0.4	0.3	0.1	0.1	(17)
(18)		PrecStd	3.1	0.4	0.3	0.4	0.8	1.5	1.6	0.9	0.7	0.7	1.3	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	51.1	62.8	68.5	76.0	84.3	87.1	85.7	81.7	72.4	61.1	50.0	(19)
(20)			MCWB	38.5	37.9	43.6	47.6	53.2	57.5	58.8	58.4	57.2	51.2	43.1	37.8	(20)
(21)		2%	DB	45.0	46.6	57.4	62.0	69.5	78.0	82.8	82.2	78.0	66.5	56.3	45.3	(21)
(22)			MCWB	34.1	36.1	41.7	45.7	51.7	58.1	59.8	57.4	55.8	47.9	41.1	34.9	(22)
(23)		5%	DB	41.9	42.2	52.2	56.9	65.8	74.6	79.7	78.9	74.9	62.4	52.6	41.7	(23)
(24)			MCWB	32.6	33.9	39.0	43.5	51.7	57.8	59.7	57.6	54.0	46.5	39.3	32.5	(24)
(25)		10%	DB	39.0	37.9	48.1	52.4	61.5	71.0	76.7	76.0	71.2	58.3	48.7	38.5	(25)
(26)			MCWB	31.0	31.0	37.1	41.1	49.3	56.7	59.5	57.1	52.4	44.8	37.4	30.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	39.4	45.9	50.1	57.8	65.7	66.7	63.9	63.5	52.9	44.3	39.0	(27)
(28)			MCDWB	51.9	48.1	60.9	64.9	67.2	75.8	76.9	73.7	74.0	67.3	59.7	49.3	(28)
(29)		2%	WB	35.3	37.0	42.6	47.0	55.0	62.2	64.4	62.1	59.3	50.4	41.6	35.7	(29)
(30)			MCDWB	43.4	46.4	57.3	60.6	64.2	72.3	74.6	72.9	70.6	62.8	54.5	43.8	(30)
(31)		5%	WB	33.2	34.5	39.9	44.6	52.8	59.9	63.0	60.7	56.6	48.0	39.8	33.5	(31)
(32)			MCDWB	40.8	42.7	51.9	56.5	62.7	71.3	73.6	72.5	69.6	60.3	51.5	41.0	(32)
(33)		10%	WB	31.5	32.1	37.7	41.9	50.6	58.0	61.3	59.2	54.7	45.7	37.4	31.4	(33)
(34)			MCDWB	38.5	38.8	47.9	52.0	60.7	69.5	73.1	72.1	67.8	56.7	47.6	39.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.3	14.9	15.2	15.9	15.3	18.5	19.6	19.9	18.9	15.4	14.5	13.5	(35)
(36)			MCDBR	15.6	15.0	18.2	21.7	19.0	22.5	21.9	21.9	21.0	19.7	17.0	14.4	(36)
(37)			MCWBR	9.7	8.7	9.8	11.4	10.1	11.7	10.3	9.8	8.9	9.2	8.8	8.3	(37)
(38)		5% WB	MCDWB	16.9	16.4	18.1	21.2	17.8	21.2	19.6	19.9	18.8	18.9	15.9	14.3	(38)
(39)			MCWBR	11.5	10.0	10.2	12.0	10.3	12.9	11.9	10.7	9.1	10.4	8.9	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.234	0.241	0.257	0.279	0.296	0.306	0.324	0.325	0.293	0.268	0.244	0.231	(40)	
(41)		taud	2.609	2.596	2.605	2.532	2.550	2.542	2.508	2.485	2.573	2.619	2.598	2.606	(41)	
(42)		Ebn at Noon	296	311	316	314	309	305	298	294	298	295	288	286	(42)	
(43)		Edh at Noon	19	23	26	31	31	32	32	32	27	22	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	585	880	1278	1574	1878	2209	2293	1970	1510	974	648	483	(44)	
(45)		RadStd	30	46	94	83	183	182	81	71	102	85	59	32	45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page